

Identifying Criterion-Related Patterns of Predictor Scores Using Multiple Regression

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Along with examples involving vocational interests and mathematics achievement, the authors describe a multiple regression based, pattern recognition procedure that can be used to identify a pattern of predictor scores associated with high scores on a criterion variable. This pattern is called the *criterion pattern*. After the criterion pattern has been identified, a second regression procedure can be used to estimate the proportion of variation attributable to the criterion pattern. Cross-validation can then be used to estimate the variation attributable to a criterion pattern derived from regression weights estimated in another sample. Finally, issues of criterion pattern invariance and interpretation are discussed.

According to Meehl (1950), *pattern* is one of the most important words in the clinician's vocabulary. He questioned whether predictor score patterns can improve criterion score prediction. In this article, we do not attempt to answer Meehl's question as to the predictive power of score patterns, but we do describe regression methods by which researchers can address the question.

In some research, the investigator may be interested in determining whether there is a pattern of predictor scores associated with high scores on a criterion variable and, if so, in determining how much variation in criterion scores can be attributed to that pattern. For the moment, we define a *profile pattern* as the arrangement of the scores in a single respondent's vector of predictor scores. If there is a predictor-score arrangement associated with high scores on the criterion, we call this arrangement the *criterion pattern*.

To better understand the research question of inter-

est, consider the hypothetical data in Table 1, loosely patterned after a study by Goldberg (1970). For 6 clinical patients, the top portion of Table 1 shows scores on four scales of a hypothetical personality inventory, the Inventory of Personality and Mood Manifestations (IPMM). The scales are Anxiety (A), Hypochondriasis (H), Schizophrenia (S), and Bipolar Disorder (B). The table also contains a dichotomous criterion variable, *N-P*, that indicates whether the patient was diagnosed as neurotic ($N-P = 1$) or psychotic ($N-P = 0$). Visually inspecting this data shows that the neurotic and psychotic patients have different score patterns. In each neurotic patient's row of data, Clients 1–3, the A and H scores tend to be higher than the S and B scores. In each psychotic patient's row of data, Clients 4–6, the A and H scores are lower than the S and B scores.

In the small sample at the top of Table 1, the pattern associated with the neurotic diagnosis can readily be discerned visually. In a larger sample, however, visual identification of the pattern can be more difficult, and some formal analysis may be needed to find the pattern associated with high scores on the criterion variable. This article describes how multiple regression can be used to identify such a pattern and how multiple regression can be used to quantify the variation accounted for by that pattern. That is, we first describe a multiple regression procedure for identification of the criterion pattern. Then we describe correlation and regression procedures that can be applied after the criterion pattern has been identified to estimate the variation accounted for by that pattern.

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Table 1

Inventory of Personality and Mood Manifestation (IPMM) Profiles of 6 Normal and 6 Clinical Respondents Diagnosed as Either Neurotic or Psychotic

Client and vector	IPMM predictor scale and variables							
	A	H	S	B	N-P	X_p	Cov_{pc}	C-N
Clinical sample								
1	75	60	50	50	1	58.75	115.4	1
2	60	75	45	55	1	58.75	150.0	1
3	60	60	55	45	1	55.00	98.1	1
4	50	50	75	60	0	58.75	-115.4	1
5	45	55	60	75	0	58.75	-150.0	1
6	55	45	60	60	0	55.00	-98.1	1
Normal sample								
7	25	40	50	50		41.25		0
8	40	25	55	45		41.25		0
9	40	40	45	55		45.00		0
10	50	50	25	40		41.25		0
11	55	45	40	25		41.25		0
12	45	55	40	40		45.00		0
Regression weights and criterion-pattern vectors								
b_v	0.00923	0.02308	-0.00923	-0.02308				
$b_v - b$	0.00923	0.02308	-0.00923	-0.02308				
$x_c: k = 500$	4.62	11.54	-4.62	-11.54				
$x_c: k = 1,000$	9.23	23.08	-9.23	-23.08				
$x_c: k = 2,000$	18.46	46.16	-18.46	-46.16				

Note. A = Anxiety; H = Hypochondriasis; S = Schizophrenia; B = Bipolar Disorder; N-P = neurotic versus psychotic criterion variable; X_p = the profile level of person p ; Cov_{pc} = the covariance between the score profile of person p and the criterion-pattern vector; C-N = clinical versus normal sample (1 = clinical, 0 = normal).

The literature contains a variety of methods for studying the patterns that appear in profiles (Games, 1990; Moskowitz & Hershberger, 2002; Nesselroade, 2001), for example, cluster analysis (Glutting, McGrath, Kamphaus, & McDermott, 1992; Gore, 2000; Lorr, 1994), configural frequency analysis (Stanton & Reynolds, 2000; von Eye, 1990; von Eye, Spiel, & Wood, 1996), modal profile analysis (Moses & Pritchard, 1995; Skinner, 1979), and multidimensional scaling (Davison, Gasser, & Ding, 1996; Davison, Kuang, & Kim, 1999). Such methods typically presume a data matrix containing V variables and P observations on each variable. Each row of the data matrix, $X_{p1}, \dots, X_{pV}$ ($p = 1, \dots, P; v = 1, \dots, V$), constitutes a profile of scores for person p .

Cluster analysis, modal profile analysis, and multidimensional scaling are internal analyses in the sense that the analysis relies solely on the V variables within each profile. No external criterion or diagnostic category variable is used to identify the prominent profile types.

In clinical practice and applied research, the fundamental question often is not whether prominent pat-

terns exist but whether the patterns have any theoretical or applied validity (Davison, 2000). Or as Meehl (1950) phrased the question, Do the patterns have any predictive efficiency? A two-step procedure for addressing this question involves first identifying frequent or prominent patterns using an internal procedure described above and then investigating the patterns' validity in a second step. Nothing in an internal procedure would guarantee that identified patterns have associations with external criterion variables. A more efficient, one-step procedure would involve identifying profile patterns on the basis of their associations with external criteria or diagnostic categories. Any pattern identified in this fashion would have associations with at least one external variable.

Given discrete predictor and criterion variables, configural frequency analysis (Stanton & Reynolds, 2000; von Eye, 1990; von Eye et al., 1996) can be used to study the relationship between configurations of scores on predictor variables and a criterion variable. That is, configural frequency analysis can be used to determine whether certain predictor-response

patterns occur more frequently in some criterion categories than in others. If the predictor (or criterion) variables are continuous, they must be artificially divided into categories before a configural frequency analysis can be performed (e.g., Stanton & Reynolds, 2000). This artificial categorization of predictor scores leads to a loss of predictor information.

Another approach is to identify all people in a given diagnostic category (or all people high on a criterion variable) and compute the mean predictor profile for the people in that diagnostic category. That mean profile is in some sense typical of the category, but it may not distinguish members of the category from members of other categories. That is, the mean profile characteristic of people in the diagnostic category may also characterize people in other categories as well.

The procedures described in this article rely on multiple regression techniques that are applicable to continuous predictor and criterion variables. We begin by showing how multiple regression can be used to identify the criterion pattern, the pattern of predictor scores associated with high scores on the criterion. After identification of the criterion pattern, correlation and regression can be used to quantify the variation accounted for by the criterion pattern. Next, we describe a cross-validation technique to estimate the shrinkage in R^2 that would result when the criterion pattern is used to predict the criterion in a new sample. Illustrative examples are presented in which correlation and regression are used to identify the criterion pattern and to determine the amount of variation accounted for by that pattern. Finally, we discuss several issues involving the invariance and the interpretation of the criterion pattern.

Identifying the Criterion Pattern

In explaining the regression procedures below, we used a dot notation to designate row means, column means, and the grand mean. The mean of the row corresponding to person p is designated as $X_{p.} = (1/V)\sum_v X_{pv}$. The mean of column v is designated as $X_{.v} = (1/P)\sum_p X_{pv}$. Similarly, the grand mean is designated as $X_{..} = (1/PV)\sum_{pv} X_{pv}$. The dot notation is also used to designate the mean of the multiple regression weights, $b_{.} = (1/V)\sum_v b_v$.

The multiple regression analysis below is based on a decomposition of each person's profile of predictor scores into two components: a level component and a pattern component. The *level component* is simply the

mean of the predictor scores for person p , $X_{p.} = (1/V)\sum_v X_{pv}$. This mean quantifies the overall elevation of the scores in person p 's profile. The *pattern component* is the V -length vector containing the deviation of each predictor score from person p 's mean: $\mathbf{x}_p = [X_{pv} - X_{p.}]$. Because the elements of the pattern vector are expressed as deviations about the person's own mean, they always sum to zero.

To illustrate the decomposition of a profile into level and pattern components, consider the data of Client 3 in Table 1 with the predictor profile vector (60, 60, 55, 45). The mean of these four scores, the level component, equals 55. The pattern vector (5, 5, 0, -10) can be computed by subtracting 55 from each element in the respondent's predictor-score vector.

When there are four predictors, as in the example of Table 1, the multiple regression equation for the expected criterion score of person p , Y'_p , can be written as $Y'_p = b_1X_{p1} + b_2X_{p2} + b_3X_{p3} + b_4X_{p4} + a$, where b_v is the regression weight for predictor variable v , and a is the intercept constant. More generally, this equation can be written for any number of V predictor variables as follows:

$$Y'_p = \sum_v b_v X_{pv} + a. \quad (1)$$

The regression weights in Equation 1 can be used to identify a criterion pattern specified as a set of pattern vectors. As we show below, these pattern vectors are called *criterion-pattern vectors*, because the regression equation for predicting the criterion can be rewritten in terms of any pattern vector in that set. All of the vectors in that set have the same arrangement of scores and, in that sense, they all have the same pattern.

Finding the Criterion Pattern

Before describing how to find a criterion pattern, we must first define it more precisely. To do so, let $\mathbf{b}^*$ be the V -length vector of least squares regression weights in Equation 1, expressed as deviations about the regression weight mean, $\mathbf{b}^* = [b_v^* = b_v - b_{.}]$. Here $b_{.}$ is the mean regression weight, $b_{.} = (1/V)\sum_v b_v$. The vector $\mathbf{b}^*$ is a member of the criterion-pattern vectors. The set of criterion-pattern vectors can be specified in terms of $\mathbf{b}^*$ as follows: A vector of scores, $\mathbf{x}_c$, belongs to the set of criterion-pattern vectors if there exists a scalar constant $k > 0$ such that

$$\mathbf{x}_c = k\mathbf{b}^*. \quad (2)$$

Because all members of the criterion pattern set are

proportional, they all share the same pattern. That is, they all have the same arrangement of highest to lowest scores. If a pattern is defined as an arrangement of scores, then each value of k specifies an exemplar of the arrangement. The pattern is the arrangement common to all exemplars.

As shown below, the regression equation can be rewritten in terms of any member of the criterion pattern set. In some cases, it suffices to specify the equation in terms of $\mathbf{b}^*$ itself (i.e., $k = 1$). In other cases, values of $k \neq 1$ facilitate graphical or tabular presentation of the pattern. In still other applications, there will be several criterion variables, each delineating a different set of criterion-pattern vectors. The researcher may then wish to compare the several patterns in terms of the degree to which they characterize the data of a given respondent. Appropriate choices of k can facilitate such comparisons. The issue of selecting k will be discussed further below.

Using the definition of the criterion-pattern vectors in Equation 2, we can use multiple regression to identify the set of pattern vectors associated with a criterion variable.

As an example, consider the three pattern vectors, $\mathbf{x}_c$, shown at the bottom of Table 1, all of which are members of the criterion set for the criterion variable $N-P$. To derive these three score vectors, we regressed the four predictors in Table 1, A , H , S , and B , onto the criterion variable, $N-P$, to obtain the four unstandardized regression weights, 0.00923, 0.02308, -0.00923, and -0.02308, for predictors A , H , S , and B , respectively. Ordinarily, one would have to subtract the mean regression weight from each of these four values to obtain the elements of $\mathbf{b}^*$. However, the mean of these four regression weights equals zero, and therefore the regression weights themselves are the elements of $\mathbf{b}^*$.

Multiplying the elements of $\mathbf{b}^*$ (the four regression weights) by $k = 500$ gave the first member of the criterion set shown at the bottom of Table 1. Multiplying by $k = 1,000$ gave the second member, and multiplying by $k = 2,000$ gave the third member. All three score vectors in Table 1 have the same arrangement of highest to lowest scores. That is, in all three of these score vectors, H is the highest score, A is the

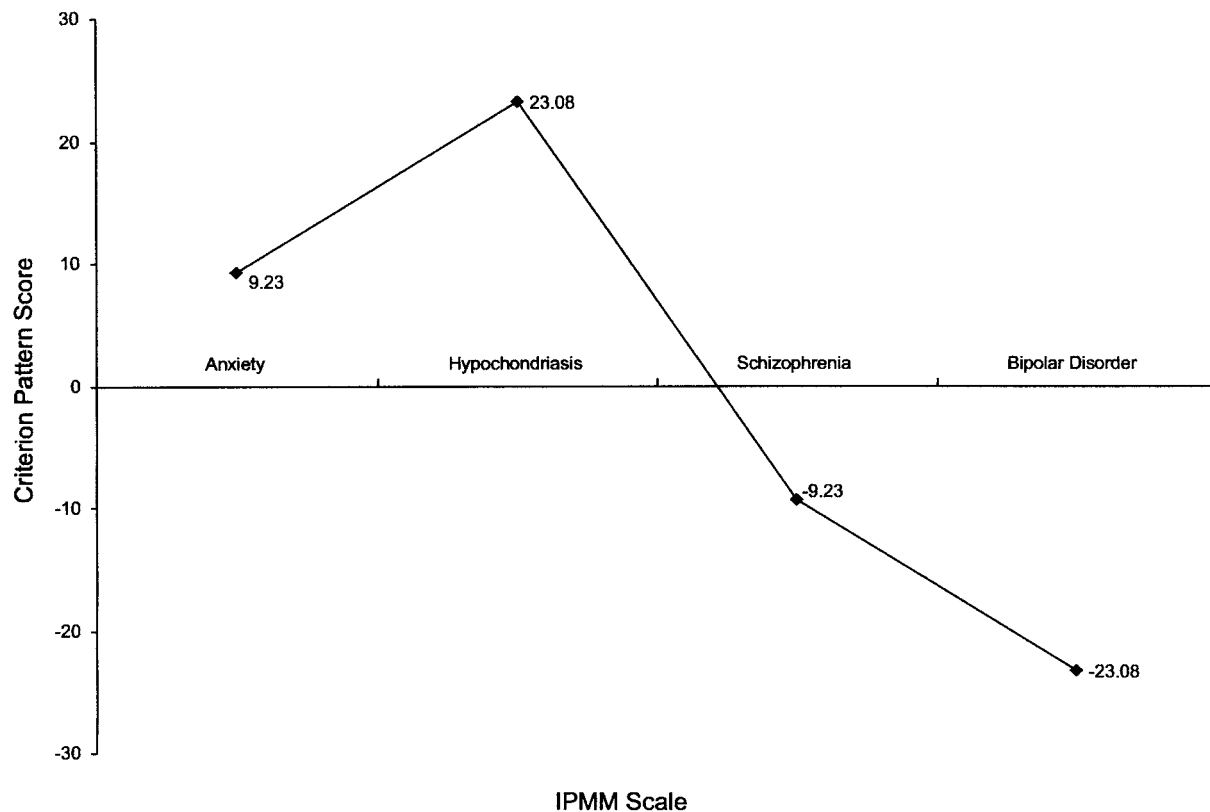


Figure 1. Criterion-pattern vector for criterion variable $N-P$ in Table 1. $k = 1,000$; IPMM = Inventory of Personality and Mood Manifestation.

second highest score, S is the third highest score, and B is the lowest score. Figure 1 shows the criterion-pattern vector for $k = 1,000$. In that score vector, A and H are the highest scores, and S and B are the lowest scores. This arrangement is consistent with the pattern observed in the patient's data: Each neurotic patient had a higher score on A and H than on S and B.

Reexpressing the Regression Equation in Terms of a Criterion-Pattern Vector

As shown in Appendix A, the regression Equation 1 can be reexpressed as

$$Y'_p = \sum_v (b_v - b) (X_{pv} - X_p) + \sum_v b X_p + a. \quad (3)$$

In Equation 3, the first sum on the right side is a linear combination of the elements in person p 's pattern vector: $(X_{pv} - X_p)$. Hereafter, this sum will be called the *pattern effect*. The second sum on the right is a function of person p 's profile level X_p . Therefore, this sum will be called the *level effect*.¹

As is further shown in Appendix A, for any criterion-pattern vector $\mathbf{x}_c$, Equation 1 (or Equation 3) can be rewritten in the form²

$$Y'_p = (V/k)Cov_{pc} + VbX_p + a, \quad (4)$$

where V is again defined as the number of variables, $Cov_{pc} = (1/V)\sum_v (X_{pv} - X_p)(X_{cv} - X_c)$ is the covariance between the actual predictor scores of person p and the scores in criterion-pattern vector $\mathbf{x}_c$, X_p is the mean of the predictor scores for person p , and X_c is the mean of the scores in $\mathbf{x}_c$.

In Equation 4, the term involving Cov_{pc} is the pattern effect. The term involving X_p is the level effect. Equation 4 shows how regression Equation 1 can be expressed in terms of any pattern vector in the criterion set. More precisely, the regression equation can be rewritten in terms of the covariance between the scores in any criterion-pattern vector $\mathbf{x}_c$ and the predictor scores of person p . For any given profile level, X_p , the predicted value Y'_p increases as its match to the criterion pattern Cov_{pc} increases. Thus, the pattern effect is conditional on profile level.

For purposes of illustration, Table 1 shows the covariance Cov_{pc} for the 6 people at the top of Table 1. Here, the criterion-pattern vector is that shown at the bottom of Table 1 for $k = 1,000$. Each of these six quantities, Cov_{pc} , is the covariance between the four predictor scores of person p and the four scores in the criterion-pattern vector for $k = 1,000$. For each of the 6 people at the top of Table 1, the mean of their scores

on the four predictors is shown in the column labeled X_p . With any multiple regression program, readers can readily verify the relationship between Equations 1 and 4 by showing that the same predicted value (within rounding error) is obtained for every respondent either by (a) regressing the criterion $N-P$ onto the four predictors A , H , S , and B , or (b) regressing the criterion variable $N-P$ onto the two predictors X_p and Cov_{pc} . Furthermore, the same multiple correlation is obtained from either of these two regressions.

It remains to determine the degrees of freedom associated with the pattern effect and the level effect. Given V predictors, there are V degrees of freedom associated with the regression weights. These degrees of freedom must be divided between the pattern effect and the level effect. The mean of the regression weights in Equation 1 consumes only one degree of freedom. The mean regression weight is the parameter associated with the level effect in Equation 4. Therefore, there is only one degree of freedom associated with the level effect. The remaining $V - 1$ degrees of freedom are associated with the pattern effect. In computing any criterion-pattern vector, V parameters ($b_v - b$) are used, but these parameters are subject to the constraint that their sum must be zero. Hence, only $V - 1$ of these parameters are free. Consequently, there are only $V - 1$ degrees of freedom associated with the pattern effect.

Estimating the Variation Accounted for by the Pattern and Level Effects

Once the criterion pattern has been identified, additional correlation and regression procedures can be used to estimate the variation accounted for by the profile pattern effect and the profile level effect. The predictors in these additional correlation and regression procedures are the two predictors in Equation 4, Cov_{pc} and X_p , rather than the V variables in Equation 1.

After the V predictor weights have been estimated using multiple regression, any criterion-pattern vector can be specified using Equation 2 and any desired value of k . Given the specification of a criterion-pattern vector, for each person one can calculate Cov_{pc} as we did in Table 1 using the criterion pattern for which $k = 1,000$. The correlation between Y and

¹ Equation 3 is Equation A3 in Appendix A.

² Equation 4 is Equation A6 in Appendix A.

Cov_{pc} can be calculated. The squared correlation will be the proportion of variation in Y that can be accounted for by the profile pattern effect alone. Because Cov_{pc} indexes the match between the actual profile of person p and the criterion-pattern vector, the squared correlation can also be interpreted as the proportion of variation in Y that is attributable to the match with the criterion profile pattern. Using the criterion variable $N-P$ and the values Cov_{pc} in Table 1, we found that the correlation was .98. In this example, 96% of the variation in the criterion variable $N-P$ can be attributed to the pattern effect alone.

Similarly, one can compute each person's profile level, X_p , and the correlation between Y and X_p . The squared correlation will be the proportion of variation in Y predicted from profile level alone. In the example of Table 1, the correlation between level and the criterion variable $N-P$ was .00. None of the variation in $N-P$ can be attributed to individual differences in profile level.

By performing a multiple regression to predict Y from the two predictors Cov_{pc} and X_p , one can estimate how much variation the pattern effect can predict over and above the level effect, how much variation the level effect can predict over and above the pattern effect, and the amount of variation accounted for by the two combined. First, consider the question of whether the pattern effect accounts for variation over and above the level effect. Let R_F^2 be the squared multiple correlation obtained when both the pattern and the level effects are included in the model. Let R_L^2 be the squared multiple correlation for the model with just the level predictor. Because the model including just the level effect is hierarchically nested within the model containing both level and pattern effects, one can then test the null hypothesis about population multiple correlations, $H_0: R_F^2 = R_L^2$, using the usual F statistic with $V - 1$ and $N - V - 1$ degrees of freedom, $F = [(R_F^2 - R_L^2)(N - V - 1)] / [(1 - R_F^2)(V - 1)]$. For our example in Equation 1, $R_F^2 = .96$, $R_L^2 = .00$, and $F(3, 1) = 8.00$, $p > .05$. Thus, in this example with a large number of predictors relative to the number of respondents, we would not reject the null hypothesis, $H_0: R_F^2 = R_L^2$, in favor of the alternative even though the pattern effect adds 96% to the variation accounted for over and above level alone.

Next, consider the question of whether the level effect accounts for variation over and above the pattern effect. Again, let R_F^2 be the squared multiple correlation obtained when both the level and pattern effects are included in the model. Let R_P^2 be the squared

multiple correlation for the model with just the pattern effect. Because the model including just the pattern effect is hierarchically nested within the model containing both level and pattern effects, one can then test the hypothesis $H_0: R_F^2 = R_P^2$ using the usual F statistic with 1 and $N - V - 1$ degrees of freedom, $F = [(R_F^2 - R_P^2)(N - V - 1)] / [(1 - R_F^2)]$. For our example in Equation 1, $R_F^2 = .96$, $R_P^2 = .96$, and $F(1, 1) = 0.00$, $p > .05$. Thus, we would not reject the null hypothesis, $H_0: R_F^2 = R_P^2$, in favor of the alternative.

To summarize the results of this example, from the regression weights we were able to identify the set of criterion-pattern vectors, three of which are shown at the bottom of Table 1 and one of which is plotted in Figure 1. As expected from our visual inspection of the data, the criterion-pattern vectors contain higher scores on the A and H scales than on the S and B scales. Correlating the profile match statistic, Cov_{pc} , led to the conclusion that the pattern effect alone accounted for 96% of the variation in the criterion $N-P$. Correlating the level statistic, X_p , with the criterion variable led to the conclusion that the level effect alone accounted for no variation in $N-P$. Together the pattern and level effects accounted for 96% of the criterion variation.

A Second Criterion Variable

To briefly extend the example in Table 1, a second criterion variable has been added for the full sample composed of both the 6 clinical patients and the 6 normal respondents. This second criterion variable, $C-N$, distinguishes between the clinical respondents ($C-N = 1$) and the normal respondents ($C-N = 0$). The last column of Table 1 shows this second criterion variable. As can be seen by inspecting the level variable, X_p , in Table 1, the clinical sample tends to have more elevated profiles than does the normal sample. Therefore, one would expect the level effect to account for a substantial portion of the variation in this second criterion variable.

First, the criterion variable $C-N$ was regressed onto the four predictors A , H , S , and B . This yielded a squared multiple correlation of .96 and four unstandardized regression weights, 0.01844, 0.01341, 0.01844, and 0.01341, for A , H , S , and B , respectively. Subtracting the mean, $b = .015925$, from each of these weights gives the criterion-pattern vector $\mathbf{b}^* = .002515, -.002515, .002515$, and $-.002515$, respectively. The correlation between the criterion and level effect is .973, which is highly significant (because of the increased sample size, which is twice that of the

first example), and the squared multiple correlation is .95. In contrast, the correlation between the criterion and the pattern effect is exactly zero. Thus, our hypothetical clinical patients are distinguished from nonpatients by their high level of symptoms. In contrast, clinical patients, neurotics, and psychotics, are distinguished from each other by their pattern of symptoms (as shown in the first example).

Cross-Validation

As noted by Dawes (1979) and others, regression weights are notoriously unstable from sample to sample. When regression weights are derived in one sample, often the regression equation will not account for as much variation in a second sample. Cross-validation, jackknifing, bootstrapping, or Browne's (1975a, 1975b) procedure can be used to estimate the drop in R^2 to be expected when the regression weights estimated in one sample are then applied in a new sample. A similar issue arises with respect to a profile pattern derived in one sample, because the profile pattern is a function of the regression weights. When a profile pattern is derived from the regression weights estimated in one sample, that pattern may not account for as much variation in a second sample. Cross-validation is one way to estimate the drop in R^2 to be expected when a profile pattern derived in one sample is then applied in a second sample.

In the real data examples below, we first applied multiple regression to the full sample to identify the criterion profile pattern and to estimate the variation accounted for by that pattern. Then, we applied cross-validation to estimate the variation that would be accounted for when a profile pattern derived from regression weights in one sample is then applied to a second sample. In this section, we describe the cross-validation procedure used in the real data examples below.

For these analyses, we randomly assigned approximately 50% of the people to Sample 1 and 50% to Sample 2. The full model was fitted in Sample 1 to estimate the regression weights. From the regression weights, we then estimated a criterion-pattern vector using the definition in Equation 2. In estimating the criterion-pattern vector, any value of k can be used, because the amount of variation accounted for by the criterion-pattern vector will be the same irrespective of k . For each person in Sample 2, we computed the covariance between their actual predictor profile scores and the scores in the criterion-pattern vector

estimated in Sample 1, Cov_{pc} . For each person in Sample 2, we also computed the level of their profile, X_p .

Having computed Cov_{pc} and X_p for each person in Sample 2, we then computed (a) the correlation between the criterion variable and X_p , (b) the correlation between the criterion variable and Cov_{pc} , and (c) the multiple correlation obtained by regressing the criterion variable onto both X_p and Cov_{pc} . This allowed us to estimate the proportion of variation accounted for by the level effect alone, the pattern effect alone, and the combination of the level effect and the pattern effect when the profile-match parameter, Cov_{pc} , is computed with respect to a criterion-pattern vector based on regression weights from a previous sample (Sample 1). In the cross-validation sample, there will be only one degree of freedom associated with the pattern effect, the degree of freedom associated with the regression weight for the predictor Cov_{pc} .

This process was then repeated reversing the roles of Samples 1 and 2. That is, Sample 2 was used to estimate the regression weights and an associated criterion-pattern vector. This criterion-pattern vector estimated in Sample 2 was then used to compute Cov_{pc} for each person in Sample 1. This profile-match statistic, Cov_{pc} , was then used to predict the criterion variable in Sample 1 both by itself and in combination with the level statistic X_p . This double-cross validation scheme provides two estimates of the variation accounted for in a second sample by a profile pattern estimated in a prior sample, one estimate based on Sample 1 serving as the cross-validation sample and another based on Sample 2 serving as the cross-validation sample.

Examples

In the examples below, our reported results emphasize the pattern effect. That is, we first show the graph of a criterion-pattern vector identified using the predictor weights from a multiple regression analysis based on the full sample. Then we show a table listing the percentage of variation accounted for by the pattern effect alone, the percentage of variation that the pattern effect accounts for over and above the level effect (called the *pattern increment*), and the total amount of variation accounted for by both pattern and level. These tables show results for the full sample and two cross-validation samples.

In evaluating each criterion pattern estimated in the full sample, we first used the significance test de-

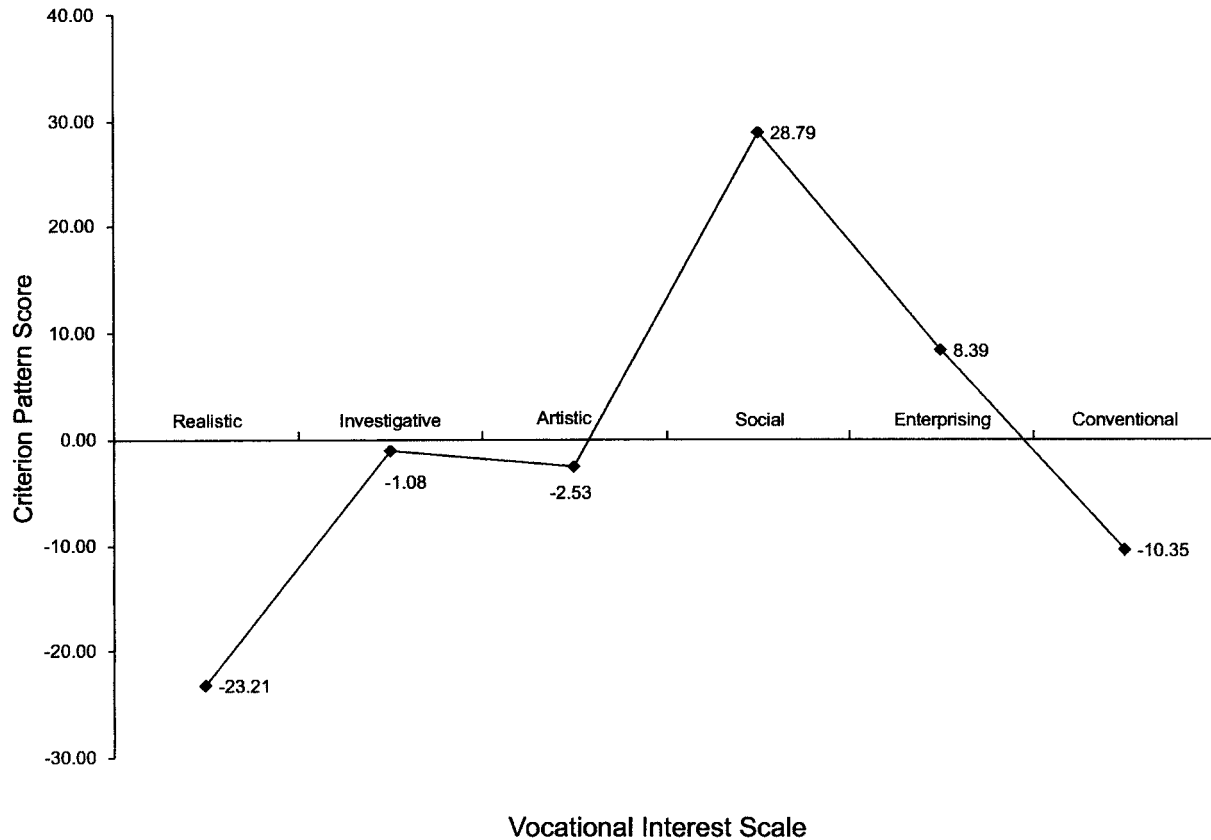


Figure 2. Criterion-pattern vector for the Social Closeness criterion of Example 1. $k = 100$.

scribed in Appendix B to determine which regression weights were significantly above the mean regression weight. That is, we tested the hypothesis $H_0: b_v = \mu(b_v)$ for each value of v . In effect, we concluded that if the regression weight, b_v , could be considered to differ significantly from the mean of the regression weights, then the corresponding element in the criterion-pattern vector, X_{cv} , could be considered to differ significantly from the mean score in the criterion-pattern vector. Interpretation of the criterion-pattern vector was based primarily on variables for which the hypothesis $H_0: b_v = \mu(b_v)$ could be rejected at the .05 level. Results of the significance test were employed to identify marker variables used to label or interpret the pattern but not to determine which variables to retain in the regression equation.

Example 1: Predicting a Personality Scale From a Vocational Interest Profile

In this example, the predictor variables were the six Holland (1973) Basic Theme scales from the Strong-Campbell Interest Inventory (Campbell & Hansen,

1985). The criterion variable was a personality scale from Tellegen's (1982) Multidimensional Personality Questionnaire, the Social Closeness scale. The respondents were 328 clients of the Minnesota Vocational Assessment Clinic.³ The researcher wanted to know whether the personality trait, social closeness, is associated with a pattern of vocational preferences reflected in the six vocational interest scales.

Full-sample results. We first estimated unstandardized regression weights in the full sample. Then we converted each regression weight to a deviation about the mean regression weight. The criterion-pattern vector for $k = 100$ is plotted in Figure 2. The null hypothesis, $H_0: b_v = \mu(b_v)$, could be rejected for only two vocational interest scales, the Social scale and the Realistic scale. Thus, the criterion-pattern vector in Figure 2 is marked by a significantly high score on the Social scale and a significantly low score

³ Our thanks to Rene Dawis, David J. Weiss, Rex Blake, and Sharon Sackett, who collected and assembled this data.

on the Realistic scale. Social occupations are generally human service professions. Realistic occupations include jobs, such as mechanic or engineer that involve making or repairing things. Someone with this interest pattern would seem to prefer working with people over working with things. The criterion pattern in Figure 2 has the same marker variables, the Social and Realistic scales, as the People versus Things dimension in Prediger's (1982) representation of Holland's (1973) hexagonal model for vocational interests. This same People versus Things dimension appears in the multidimensional scaling studies (Davison, 1994; Rounds, Davison, & Dawis, 1979) of correlations among vocational-interest scales.

In this example, the correlation between interest profile level and Social Closeness scores was only .08. The correlation with the pattern effect was somewhat higher, .36, suggesting that the pattern effect alone accounted for 12.9% of the variation in Social Closeness scores. The level effect and the pattern effect were nearly uncorrelated, .04.

The pattern effect added a statistically significant 12.7% to the variation accounted for by the level effect alone. Together, the level effect and the pattern effect accounted for only slightly more than 13% of the variation in Social Closeness scores.

Cross-validation results. The cross-validation results largely confirm the full-sample findings in the sense that, on cross-validation, the pattern effect, not level effect, accounted for most of the predictable variation in the criterion. In Sample 1, level accounted for less than 1% of the variation in Social Closeness scores. The criterion-pattern vector identified from regression weights in Sample 2 accounted for 6.7% of the criterion variation in Sample 1. That criterion pattern added an additional 6.4% to the variation accounted for over and above level. Together, level and pattern accounted for 7.1% of the variation in the Social Closeness criterion variable.

In Sample 2, level again accounted for less than 1% of the variation in Social Closeness scores. The criterion-pattern vector identified from regression weights in Sample 1 accounted for 12.2% of the criterion variation in Sample 2. The criterion pattern found in Sample 2 added an additional 11.8% to the variation accounted for over and above level.

The results from the full-sample and the cross-validation samples lead to similar conclusions. Scores on the six vocational-interest scales account for only a small portion of the variation in Social Closeness scores. Profile pattern, not level, accounts for most of

the predictable variation. The data support an association between this personality characteristic and a vocational interest pattern in which persons high on Social Closeness tend to prefer social occupations over realistic ones.

Example 2: Predicting High School Mathematics Achievement From High School Course Work

Our second example does not involve a test-battery profile. Rather, it involves predicting a high school student's senior-year mathematics achievement score from the profile of the student's high school mathematics course work as reported on his or her transcript (Davenport et al., 1998). In this case, *level* refers to the amount of course work taken in mathematics. *Pattern* refers to the distribution of the course work over various course work categories.

This example was chosen because it illustrates how the analysis described above can be applied to sets of predictor variables that, taken together, describe a frequency distribution for each respondent. In such applications, each predictor variable, X_{pv} , shows the frequency with which responses falling in category v occurred for person p . Using the analysis described above, multiple regression can be used to describe a distributional pattern associated with high scores on the criterion, and it can be used to quantify the amount of variation accounted for by that pattern.

The data for this example were taken from the National Educational Longitudinal Study (NELS; National Center for Education Statistics—U.S. Department of Education, 1990). The sample consists of 10,245 U.S. high school students who were 9th–12th graders in the years 1988–1992. The data set includes each student's high school course transcript. In prior research, mathematics courses were classified into eight categories, including a miscellaneous category. For purposes of this illustration, the miscellaneous category was dropped. The remaining seven categories are listed along the horizontal axis in Figure 3. The categories are arrayed roughly in terms of complexity from functional courses (i.e., special education) at one extreme to advanced courses (e.g., calculus, pre-calculus, analysis) at the other.

Several of the categories deserve some explanation. The basic and general mathematics courses cover general arithmetic and its application. These courses would usually be taken by students prior to Algebra 1 or by students who do not take Algebra 1. The category, Algebra, 2 Parts, refers to a 2-year course in

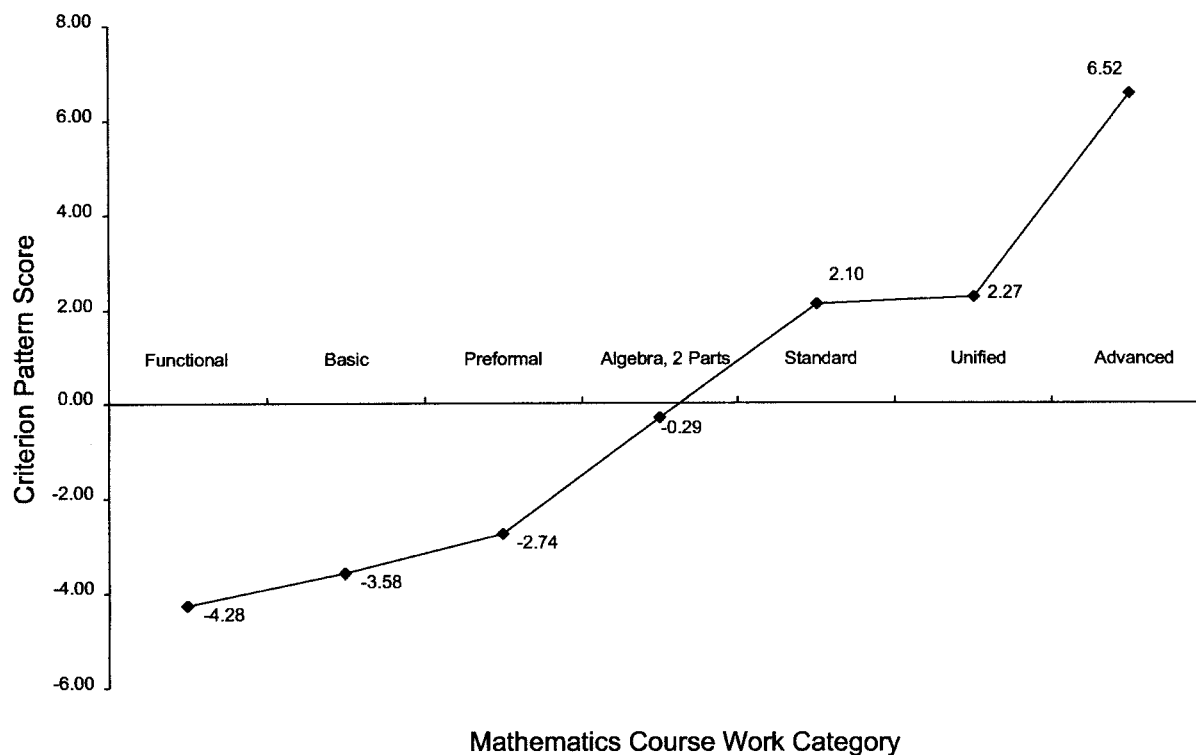


Figure 3. Criterion-pattern vector for the Mathematics Achievement criterion of Example 2. $k = 1$.

which students study Algebra 1 over a 2-year period. Such courses are designed for students who need to study algebra at a slower pace. The *standard* category refers to the standard precollege sequence of Algebra 1, Algebra 2, and Geometry. The *unified* category consists of a sequence in which students study algebra and geometry in a more integrated fashion than in the *standard* sequence. The *advanced* category generally consists of courses that would be taken after completing the *standard* or *unified* sequence.

For each student, we computed the number of Carnegie Units (CUs) completed by the student in each category. A CU equals roughly a year's worth of study or its equivalent. For most students who complete high school in 4 years with no summer school and take only one math course at a time, the number of CUs will range between 0 and 4. Thus, each student's course work profile is a set of seven scores each showing the number of CUs completed in one of the seven categories.

Full-sample results. The criterion variable, mathematics achievement, was regressed onto the seven-course-category variables. The unstandardized regression weights were converted to deviations about the regression weight mean. Then the criterion pattern for

$k = 1$ was plotted in Figure 3. The null hypothesis, $H_0: b_v = \mu(b_v)$, was rejected for all but the Algebra, 2 Parts course category.

The correlation between profile level (total CUs completed) and achievement scores was .45. The correlation with pattern (or distribution of course work over the categories) was .76, indicating that the pattern effect accounted for 57.4% of the mathematics-achievement variation. In contrast to our earlier examples (see Tables 1 and 2), the correlation between level and pattern was high, .61. That is, there is a substantial association between the amount of course

Table 2
Variation and Incremental Variation Accounted for in Social Closeness by the Vocational Interest Criterion-Pattern Vector of Figure 2: Full and Cross-Validation Samples

Percent variation accounted for	Full sample	Cross-validation	
		Sample 1	Sample 2
Pattern alone	12.9*	6.7*	12.2*
Pattern increment	12.7*	6.4*	11.8*
Level and pattern	13.3*	7.1*	12.4*

* $p < .01$.

work taken and the distribution of that course work over the course categories. Students taking more course work tend to have taken advanced course work.

As shown in Table 3, the pattern effect added an additional 36.8% to the variation accounted for by level alone. In total, the pattern and level effects accounted for 57.4% of the variation in mathematics achievement. Although level accounted for a substantial amount of the achievement variation, 21%, pattern added substantially to that amount.

Cross-validation results. Again, the cross-validation results in Table 3 confirm the findings from the full sample. In Sample 1, the pattern effect alone accounted for 58.1% of the variation in achievement. The criterion-pattern vector identified in Sample 2 accounted for an additional 36.4% of the variation over and above level alone. Together, pattern and level accounted for 58.1% of the variation in mathematics achievement.

In Sample 2, the pattern effect accounted for 56.7% of the variation in achievement. The criterion-pattern vector identified in Sample 1 accounted for an additional 37.2% of the variation over and above level alone. Together, the level and pattern effects accounted for a total of 56.8% of the variation.

These data suggest that level, or amount of course work taken, accounts for a substantial proportion of the variation in mathematics achievement, 20%. However, the pattern, or distribution of course work over the categories, more than doubles the variation accounted for. There is a very substantial correlation between amount and content of course work, if content is taken to mean the distribution of the course work over our seven categories.

In drawing graphs of the criterion patterns in Figures 1–2, we used values of k other than 1.0 for a rather trivial reason: to get rid of nuisance zeros to the right of the decimal place in the numbers being

graphed. However, there can be more substantial reasons for choosing $k \neq 1.0$. In some of our work (not reported here), we have had several criterion groups, have used multiple regression to identify criterion patterns associated with each group, and have then classified respondents into groups on the basis of the criterion pattern best matched by each respondent's data. In this research, we computed several covariance profile-match statistics, Cov_{pc} ($c = 1, \dots, C$), for each person and then classified person p into the group corresponding to their highest profile match statistic. Before computing the profile-match statistics on which classification is based, we standardized the variability within each criterion pattern by selecting a different value of k for each criterion pattern so that $(1/V)\sum x_{cv}^2 = 1.0$ for each pattern. This standardization gives profile-match statistics that are comparable across the several criterion patterns. From our experience, the choice of k seems most critical when the profile match statistic, Cov_{pc} , is used to classify individuals by examining their match to one of several criterion patterns.

Issues of Interpretation and Invariance

There are a number of issues involving either the interpretation of the multiple regression results, the invariance of results, or both. Here we begin by considering the interpretation of the analysis with respect to the equal weighting issue in multiple regression. Then we turn to the interpretation and the invariance of results with respect to various ways of scoring the predictor variables.

Unit Weighting of Predictor Variables

There is an extensive literature comparing the variation accounted for using optimal (least squares) predictor weights with the variation accounted for when equal weights are used for all predictors (e.g., Dawes, 1979; Dawes & Corrigan, 1974; Green, 1977; Wainer, 1976). When all variables are rescored, if necessary, to be positively related to the criterion, equal weights often do as well on cross-validation as do the optimal weights, particularly if the optimal weights have been estimated in a small sample. In the analysis above, the model including only the level effect is the *equal weight model*. The model including all of the predictors (which is also the model including both pattern and level effects) is the *optimal weight model*. The difference between R_F^2 and R_L^2 is the additional proportion of variation accounted for by

Table 3
Variation and Incremental Variation Accounted for in Mathematics Achievement by the Course Work Criterion-Pattern Vector of Figure 3: Full and Cross-Validation Samples

Percent variation accounted for	Full sample	Cross-validation	
		Sample 1	Sample 2
Pattern alone	57.4*	58.1*	56.7*
Pattern increment	36.8*	36.4*	37.2*
Level and pattern	57.4*	58.1*	56.8*

* $p < .01$.

using optimal weights instead of equal weights. If the null hypothesis $H_0: R_F^2 = R_L^2$ is rejected, then the equal weight model can be rejected in favor of the model that includes the optimal regression weights. Therefore, the analysis described above can be used to investigate whether optimal weights can account for more variation than equal regression weights.

Scoring of Predictor Variables

Three scoring issues can affect results and conclusions reached with respect to profile patterns: standardizing predictor means, standardizing predictor variances, and reverse scoring some predictors.

Standardizing predictor means. Paradoxically, when the means of the predictor variables are different for each variable, there can be no respondent whose data set seems to resemble the criterion pattern, even if the criterion pattern accounts for most of the criterion variation. The top of Table 4 shows such a data set. If one regresses the four predictors, *A*, *H*, *S*, and *B*, onto the criterion variable *N-P*, the set of criterion-pattern vectors identified on the basis of the regression weights is exactly the same as for the data at the top of Table 1. One of the criterion-pattern vectors is plotted in Figure 1. The criterion-pattern vector in Figure 1 accounts for 96% of the variation in the criterion variable *N-P* in Table 4. Yet no respondent's data set looks like the pattern in Figure 1.

The major feature of the criterion-pattern vector in Figure 1 is that it has higher scores for scales *A* and *H* than for *S* and *B*. None of the respondents at the top of Table 4 has a higher score on *A* and *H* than on *S* and *B*. Respondent 2's data set has the highest covariance with the criterion pattern, and yet even that response vector does not have higher scores on *A* and *H*. How can this paradox arise?

The raw score of any individual can be divided into two components: $X_{pv} = x_{pv} + X_{.v}$. Here x_{pv} is the deviation score $x_{pv} = (X_{pv} - X_{.v})$, and $X_{.v}$ is the mean of variable *v*. As shown in Appendix C, the covariance between the actual profile of person *p* and any criterion-pattern vector depends on two factors: the degree to which the pattern of person *p*'s deviation scores matches the criterion-pattern vector and the degree to which the pattern of the variable means matches the criterion pattern. Individual differences in the profile-match statistic, Cov_{pc} , are completely determined by the first of these two factors, because the second factor is a constant for all respondents (see Appendix C).

At the top of Table 4, the means of the variables *A* and *H* are so much lower than the means of *S* and *B* that no respondent has higher scores on *A* and *H* than on *S* and *B*. The pattern of the means is masking the fact that some respondents' deviation scores are higher on *A* and *H* than on *S* and *B*. As a result, the

Table 4
Raw Score and Deviation Score Profiles of 6 Hypothetical Clients Diagnosed as Either
Neurotic or Psychotic

Client	IPMM scale and variable						Cov_{pc}
	A	H	S	B	$N-P$	X_{p_i}	
Raw score							
1	90	60	90	80	1	80.00	-115.4
2	75	75	85	85	1	80.00	-80.8
3	75	60	95	75	1	76.25	-132.7
4	65	50	115	90	0	80.00	-346.2
5	60	55	100	105	0	80.00	-380.8
6	70	45	100	90	0	76.25	-328.9
Deviation score							
1	17.50	2.50	-7.50	-7.50	1	1.25	115.4
2	2.50	17.50	-12.50	-2.50	1	1.25	150.0
3	2.50	2.50	-2.50	-12.50	1	-2.50	98.1
4	-7.50	-7.50	17.50	2.50	0	1.25	-115.4
5	-12.50	-2.50	2.50	17.50	0	1.25	-150.0
6	-2.50	-12.50	2.50	2.50	0	-2.50	-98.1

Note. IPMM = Inventory of Personality and Mood Manifestations; A = Anxiety; H = Hypochondriasis; S = Schizophrenia; B = Bipolar Disorder; N-P = neurotic versus psychotic criterion variable (1 = neurotic, 0 = psychotic); $X_{.p}$ = the profile level of person *p*; Cov_{pc} = the covariance between the score profile of person *p* and the criterion-pattern vector.

covariance of every client is negative, indicating that no client has a raw-score pattern closely resembling the criterion pattern. However, all of the neurotics have higher covariances than do the psychotics, and therefore the profile-match statistic, Cov_{pc} , is a good predictor of the criterion variable.

If the variables at the top of Table 4 are converted to deviation score form, one can readily see that the neurotics' deviation scores do resemble the pattern in Figure 1. The scores expressed in deviation-score form are given at the bottom of Table 4. At the bottom of Table 4, neurotics have score patterns similar to the criterion pattern in that *A* and *H* are higher than *S* and *B*. The top of Table 1 shows the same variables standardized to have equal means of 57.5. When the variables have equal means as at the top of Table 1, it is readily seen that *A* and *H* are higher than *S* and *B* for neurotics, whereas the psychotics have the mirror-image pattern.

In effect, multiple regression can discern the fact that there is a criterion pattern such that people with high criterion scores have predictor-deviation score patterns more closely resembling that criterion pattern than do people with low scores on the criterion. Multiple regression can do so even when the predictor variables are expressed in raw-score form with unequal means. Most human observers cannot. That is, when the predictor variables are expressed as raw scores with unequal means, human observers cannot always discern the deviation-score pattern associated with high scores on the predictor variable simply by inspecting the data. Human observers cannot identify respondents whose deviation-score pattern looks like the criterion pattern. Because the important results are unaffected by standardizing means, there is no reason to standardize means for purposes of statistical analysis. However, if researchers want to visually inspect their data to identify a criterion pattern, or if researchers want to visually compare respondent scores to the criterion pattern, they should first convert the raw scores to deviation-score form, as shown at the bottom of Table 4, or the researchers should standardize the variables so all variables have equal means, as shown at the top of Table 1.

Standardizing predictor variances. Although a human observer might be influenced by unequal means, multiple regression is not. Multiple regression is, however, affected by unequal variances. Multiple regression produces both standardized and unstandardized regression weights. In the examples above, the criterion-pattern vectors have been based on un-

standardized weights. Unstandardized weights serve to identify a pattern of predictor-deviation scores associated with high scores on the criterion variable. Standardized predictor weights serve to identify a pattern of *z* scores associated with high scores on the criterion variable. That is, the standardized weights refer to predictor variables that have been standardized to have equal means and variances.

The set of criterion-pattern vectors identified using unstandardized weights may look quite different from the set of criterion-pattern vectors identified using standardized weights. That is, the arrangement of highest to lowest scores in the set of criterion vectors computed from unstandardized regression weights may be quite different from the arrangement of highest to lowest scores in the set of criterion vectors computed from standardized regression weights. If the criterion-pattern vector is computed from standardized regression weights, the predictor variables must first be standardized to have equal means and variances before the covariance (Cov_{pc}) and the level (X_p) variables are computed for each respondent.

When raw scores are expressed in meaningful units that are comparable across variables (e.g., high school CUs) or the literature focuses on patterns expressed in unstandardized units (e.g., Butcher & Williams, 2000; McAllister, 1986), researchers will want to identify patterns in the unstandardized variables using the unstandardized regression weights. In the discussion above, variables are considered standardized only if they have a mean of 0 and a variance of 1.0 in the research sample. Variables that have been standardized with respect to a norm group, but not the research sample, are considered unstandardized.

Reverse-scoring predictor variables. In the literature on unit weighting cited above, the researchers conclude that unit weights often perform as well as optimal weights on cross-validation if the predictor variables are first scored so that all have a positive correlation with the criterion variable. In that literature, some predictors are reverse scored for purposes of developing a unit weight model. In studying profile patterns, should some variables be reverse scored, if necessary, so as to be positively correlated with the criterion variable?

The answer to this question is important to the analyses described above, because reverse scoring some predictors can change the conclusions reached. For instance, in the data at the top of Table 1, the multiple regression analysis led to the conclusion that profile level accounted for no variation in the criterion

variable. Profile pattern accounted for 96% of the variation. Although we have not reported the analysis here, after reverse scoring variables *S* and *B* so that they are positively correlated with the criterion variable, the total amount of variation accounted for remains the same. However, profile level accounts for most of the variation in the criterion variable. The variation accounted for by the pattern effect is greatly reduced. Because reverse scoring of some variables can greatly change the conclusions reached, reverse scoring is an important issue in research on profile patterns.

If the goal of one's research is simply to produce a linear regression equation that will perform well in other samples, then reverse scoring some predictor variables may do no harm. In the type of research described above, however, the goal is not mere prediction. Rather the goal is to develop an understanding of the predictor profile pattern associated with high scores on the criterion variable. Developing that understanding requires a formal model expressed in terms of pattern and level effects. Consider two hypothetical studies that might produce data similar to that at the top of Table 1.

The first is a study of the clinical judgment process similar to research described by Goldberg (1970). In this hypothetical research study, a clinician is asked to view the predictor scores of all 6 clients at the top of Table 1 and then to classify the client as either neurotic or psychotic. The goal is to build a linear model of the clinician's judgment process. Presumably, the investigator would present each client's IPMM profile to the clinician with variables scored in a manner with which the clinician is familiar, even if some of the variables are then negatively correlated with the criterion variable. Furthermore, the variables entered into the model would be scored just as they were scored in the information given to the clinician. To reverse score some variables in the model would not yield a model of the clinician's judgment process, because the information in the model would not be the same as the information given to the clinician. As this example illustrates, it is not always sensible to reverse score the predictor information if the goal is to build a psychological model of human responses.

Consider a second possible study yielding information similar to that in Table 1. The 6 clients each take the IPMM on admission to the clinician's care. Without seeing the predictor variables, the clinician makes an independent diagnosis of each client as either neurotic or psychotic. The goal is to find an empirically

derived criterion pattern that subsequent clinicians can use as a guide in making diagnoses based on IPMM information. Presumably, the empirically derived pattern should include predictor information scored in the same form that the information is commonly reported to clinicians, even if some of the predictor variables are then scored so as to be negatively related to the criterion. Reverse scoring some predictors would yield a profile of no use to the clinician because it would be based on variables scored in a manner other than the scoring used in clinical score reports of actual clients. Reverse scoring some variables to make them positively correlated with the criterion variable would inhibit development of a score pattern useful to clinical practice.

In the absence of other considerations, researchers may wish to rescore predictor variables so they have positive correlations with the criterion variable. With the possible exception of research on unit weighting, however, there will usually be other considerations. In some cases, one particular direction of scoring will have been most commonly used in the prior literature, and therefore the researcher will want to adopt the common scoring so results of the new research are more easily related to the prior literature. In other cases, theory assumes a particular direction of scoring. A theory about anxiety would seem to imply variables scored so that high scores mean high levels of anxiety not a lack of anxiety. When the goal is formal modeling to enhance theory, common research practice or the theory itself may imply a direction of scoring the variables.

Discussion

Interpreting profile patterns implicitly involves comparing scores across continuous variables. Such comparisons are legitimate only if the scores from the variables are, in some sense, comparable. In our course-taking example above, for instance, the scores for each category are all expressed as CUs, a unit roughly equivalent to a year's worth of study. For our interest-inventory example, the scores were *T* scores scaled to have $M = 50$ and $SD = 10$ in the norm group. It is not our intention to defend the equivalence of CUs across high school course categories or the equivalence of *T* scores across different psychological scales. Rather, we merely want to point out that interpreting profile patterns involves comparing variables. For such comparisons to be meaningful, the variables must be in comparable units.

When the predictor variables are in comparable units and the regression of the variables on the criterion is linear, the regression weights sketch a set of criterion-pattern vectors. All of the criterion-pattern vectors are proportional to each other and, taken together, this set constitutes all of the vectors displaying a particular arrangement of the predictor variables. For a given profile level, the expected criterion score increases as the match to any criterion-pattern vector increases when match is indexed as the covariance between an observed-predictor score profile and the criterion-pattern vector. The expected criterion score can be expressed as a function of just two predictor profile effects: profile level and match to the criterion pattern. As a consequence, the accounted for variation in *Y* can be divided among two nonorthogonal sources, a profile-level effect and a profile-pattern effect. Multiple regression can be used both to identify the criterion pattern and to determine the amount of variation predictable from the match to the criterion pattern.

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Appendix A

Expressing Regression Equations in Terms of Pattern and Level Effects

The purpose of this appendix is to show the equivalence of Equations 1, 3, and 4. Let $b_v = (1/V)\sum_v b_v$ and let $X_p = (1/V)\sum_v X_{pv}$. Given these definitions, Equation 1 can be re-expressed in terms of $b_v - b$ and $X_{pv} - X_p$:

$$Y'_p = \sum_v b_v X_{pv} + a \quad (\text{A1})$$

$$= \sum_v (b_v - b + b) (X_{pv} - X_p + X_p) + a \quad (\text{A2})$$

$$= \sum_v (b_v - b) (X_{pv} - X_p) + \sum_v b X_p + a \quad (\text{A3})$$

$$= \sum_v (b_v - b) (X_{pv} - X_p) + Vb X_p + a. \quad (\text{A4})$$

Equation A3 follows from A2 because $\sum_v (b_v - b) = \sum_v (X_{pv} - X_p) = 0$. Because $\sum_v b X_p = Vb X_p$, Equation A4 follows from Equation A3. From the definition of a criterion-pattern vector (Equation 2), $(b_v - b) = (1/k) (X_{cv} - X_c)$, where X_{cv} is the v th element of the criterion-pattern vector $\mathbf{x}_c$, and X_c is the mean of the elements in the criterion-pattern vector. Using this equality, $(b_v - b) = (1/k) (X_{cv} - X_c)$, in Equation A4 yields

$$Y'_p = \sum_v (1/k) (X_{cv} - X_c) (X_{pv} - X_p) + Vb X_p + a \quad (\text{A5})$$

$$= (V/k) \text{Cov}_{pc} + Vb X_p + a. \quad (\text{A6})$$

Equation A6 follows from the definition of the covariance $\text{Cov}_{pc} = (1/V) \sum_v (X_{cv} - X_c) (X_{pv} - X_p)$. Equations A1, A3, and A6 give the desired result.

Appendix B

Statistical Test to Assist in Criterion Pattern Interpretation

To aid in the interpretation of the criterion-pattern vectors in Examples 1 and 2, we performed an approximate test of the null hypothesis $H_0: b_v = \mu(b_v)$ separately for each variable v , where $\mu(b_v)$ is the expected value of b_v over all possible random samples from this population. Ideally, the test statistic would be $t = [b_v - \mu(b_v)]/s(b_v)$, which would be distributed as Student's t with 1 and $(N - V - 1)$ degrees of freedom. Here $s(b_v)$ is the standard error of the regression

weight b_v , and $s(b_v)$ is estimated in the regression analysis. Because the population mean, $\mu(b_v)$, is unknown, we substituted the sample mean b in the computation of the test statistic. If the regression weight b_v differed significantly from the mean of the regression weights, then the corresponding element in the criterion-pattern vector, X_{cv} , would differ significantly from the mean score in the criterion-pattern vector because elements of the criterion-pattern vec-

tor are proportional to regression weights expressed as deviations about their mean. Our interpretation of the criterion-pattern vectors in Figures 2 and 3 is based on the

significant coefficients for which the null hypothesis $H_0: b_v = \mu(b_v)$ has been rejected.

Appendix C

Raw Score and Deviation Score Profile-Match Statistics

The purpose of this appendix is to show that the covariance between the actual raw score profile of person p and any criterion-pattern vector depends on two factors: the degree to which the pattern of person p 's deviation scores matches the criterion-pattern vector and the degree to which the pattern of the variable means matches the criterion pattern. Individual differences in the profile-match statistic are completely determined by the first of these two factors, because the second factor is a constant for all respondents.

Let COV_{pc} refer to the covariance between the raw scores of person p and the scores in criterion-pattern vector c . Let cov_{pc} refer to the covariance between the deviation scores of person p and criterion pattern c . Then $COV_{pc} = (1/V)\sum_v(X_{cv} - X_{c.})(X_{pv} - X_{p.})$. In this expression, the variable $X_{p.}$ is the mean raw score in the profile of person p , and $X_{c.}$ is the mean score in the criterion-pattern vector. If we let x_{pv} be the deviation score, $x_{pv} = (X_{pv} - X_{v.})$, where $X_{v.}$ is the mean of variable v , then

$$X_{p.} = (1/V)\sum_v X_{pv} = (1/V)\sum_v(x_{pv} + X_{v.}) \quad (C1)$$

$$= (1/V)\sum_v x_{pv} + (1/V)\sum_v X_{v.} \quad (C2)$$

$$= x_{p.} + X_{..} \quad (C3)$$

Here $X_{..}$ is the grand mean of raw scores. Using the result in C3 and the definition of a deviation score,

$$COV_{pc} = (1/V)\sum_v(X_{cv} - X_{c.})(X_{pv} - X_{p.}) \quad (C4)$$

$$= (1/V)\sum_v(X_{cv} - X_{c.})(x_{pv} + X_{v.} - x_{p.} - X_{..}) \quad (C5)$$

$$= (1/V)\sum_v(X_{cv} - X_{c.})(x_{pv} - x_{p.}) + (1/V)\sum_v(X_{cv} - X_{c.})(X_{v.} - X_{..}) \quad (C6)$$

$$= cov_{pc} + h, \quad (C7)$$

where $h = (1/V)\sum_v(X_{cv} - X_{c.})(X_{v.} - X_{..})$. Note that h is a constant for every person. Equation C6 shows that the COV_{pc} is a function of two terms, each corresponding to a sum on the right-hand side. The first term is equal to the covariance between the criterion-pattern vector scores and the deviation scores of person p . The second is equal to the covariance between the criterion-pattern vector scores and the vector of variable means. Equation C7 shows that all individual differences in COV_{pc} are attributable to individual differences in predictor deviation scores, that is, individual differences in cov_{pc} , because h is a constant for all people. The match parameter cov_{pc} indexes the match between the deviation scores of person p and the criterion profile pattern vector.

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